

Least Squares Regression

Nonlinear Model

Data Set:

$$(x_i, y_i) | i = 0, 1, 2, \dots, n$$

Nonlinear Model:

$$y(x) = f(x, \mathbf{a}) = f(x, a_0, a_1, \dots, a_m); \quad m < n$$

Error:

$$e_i = y_i - y(x_i) = y_i - f(x_i, \mathbf{a}) = y_i - f(x_i, a_0, a_1, \dots, a_m); \quad m < n$$

Sum of Error Squares:

$$S_e = \sum_{i=0}^n e_i^2 = \sum_{i=0}^n [y_i - f(x_i, a_0, a_1, \dots, a_m)]^2; \quad m < n$$

Minimize Sum of Error Squares:

$$\frac{\partial S_e}{\partial a_0} = 0, \quad \frac{\partial S_e}{\partial a_1} = 0, \quad \dots, \quad \frac{\partial S_e}{\partial a_m} = 0; \quad m < n$$

$$\sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_0} \right) = 0$$

This leads to a **system of $m + 1$ equations** for finding $a_0, a_1, \dots, a_m$.

$$\sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_1} \right) = 0$$

$\vdots$

$$\sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_m} \right) = 0$$

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Vector Formulation:

$$\left. \begin{aligned} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_0} \right) &= 0 \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_1} \right) &= 0 \\ &\vdots \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_m} \right) &= 0 \end{aligned} \right\} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

Approximation:

$$\sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0} \quad \rightarrow \quad \sum_{i=0}^n [y_i - f(x_i, \mathbf{a} + \Delta \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

Initial Guess
 $\downarrow$
Search Direction $\rightarrow$ Recurrence Relation

$$f(x, \mathbf{a} + \Delta \mathbf{a}) \cong f(x, \mathbf{a}) + \nabla f \cdot \Delta \mathbf{a}$$

$$\sum_{i=0}^n [y_i - f(x_i, \mathbf{a}) - \nabla f \cdot \Delta \mathbf{a}] \cdot (-\nabla f) = \mathbf{0}$$

$$\mathbf{a}^{(k)} = \mathbf{a}^{(k-1)} + \Delta \mathbf{a}^{(k)}$$

$$(\mathbf{a} \otimes \mathbf{b}) \mathbf{c} = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$$

$$\sum_{i=0}^n (\nabla f \cdot \Delta \mathbf{a}) \cdot \nabla f = \sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot \nabla f$$

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Vector Formulation:

$$\left. \begin{aligned} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_0} \right) &= 0 \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_1} \right) &= 0 \\ &\vdots \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_m} \right) &= 0 \end{aligned} \right\} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

Approximation:

$$\sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0} \quad \rightarrow \quad \sum_{i=0}^n [y_i - f(x_i, \mathbf{a} + \Delta \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

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$$\sum_{i=0}^n [y_i - f(x_i, \mathbf{a}) - \nabla f \cdot \Delta \mathbf{a}] \cdot (-\nabla f) = \mathbf{0}$$

$$\mathbf{a}^{(k)} = \mathbf{a}^{(k-1)} + \Delta \mathbf{a}^{(k)}$$

$$(\mathbf{a} \otimes \mathbf{b}) \mathbf{c} = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$$

$$\sum_{i=0}^n (\nabla f \otimes \nabla f) \cdot \Delta \mathbf{a} = \sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot \nabla f$$

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Vector Formulation:

$$\left. \begin{aligned} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_0} \right) &= 0 \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_1} \right) &= 0 \\ &\vdots \\ \sum_{i=0}^n 2 \cdot [y_i - f(x_i, a_0, a_1, \dots, a_m)] \cdot \left(-\frac{\partial f}{\partial a_m} \right) &= 0 \end{aligned} \right\} \sum_{i=0}^n 2 \cdot [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

Approximation:

$$\sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0} \quad \rightarrow \quad \sum_{i=0}^n [y_i - f(x_i, \underset{\substack{\text{Initial Guess} \\ \downarrow}}{\mathbf{a}} + \underset{\substack{\uparrow \\ \text{Search Direction}}}{\Delta \mathbf{a}})] \cdot [-\nabla f(x_i, \mathbf{a})] = \mathbf{0}$$

→ Recurrence Relation

$$f(x, \mathbf{a} + \Delta \mathbf{a}) \cong f(x, \mathbf{a}) + \nabla f \cdot \Delta \mathbf{a}$$

$$(\mathbf{a} \otimes \mathbf{b}) \mathbf{c} = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$$

$$\begin{bmatrix} \sum_{i=0}^n \frac{\partial f}{\partial a_0} \cdot \frac{\partial f}{\partial a_0} & \sum_{i=0}^n \frac{\partial f}{\partial a_0} \cdot \frac{\partial f}{\partial a_1} & \dots & \sum_{i=0}^n \frac{\partial f}{\partial a_0} \cdot \frac{\partial f}{\partial a_m} \\ \sum_{i=0}^n \frac{\partial f}{\partial a_1} \cdot \frac{\partial f}{\partial a_0} & \sum_{i=0}^n \frac{\partial f}{\partial a_1} \cdot \frac{\partial f}{\partial a_1} & \dots & \sum_{i=0}^n \frac{\partial f}{\partial a_1} \cdot \frac{\partial f}{\partial a_m} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=0}^n \frac{\partial f}{\partial a_m} \cdot \frac{\partial f}{\partial a_0} & \sum_{i=0}^n \frac{\partial f}{\partial a_m} \cdot \frac{\partial f}{\partial a_1} & \dots & \sum_{i=0}^n \frac{\partial f}{\partial a_m} \cdot \frac{\partial f}{\partial a_m} \end{bmatrix} \begin{Bmatrix} \Delta a_0 \\ \Delta a_1 \\ \vdots \\ \Delta a_m \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot \frac{\partial f}{\partial a_0} \\ \sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot \frac{\partial f}{\partial a_1} \\ \vdots \\ \sum_{i=0}^n [y_i - f(x_i, \mathbf{a})] \cdot \frac{\partial f}{\partial a_m} \end{Bmatrix} \quad \mathbf{a}^{(k)} = \mathbf{a}^{(k-1)} + \Delta \mathbf{a}^{(k)}$$

Least Squares Regression

Example 04-08: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	1	2.2
1	2	2.8
2	3	3.6
3	4	4.5
4	5	5.5

$$y(x) = f(x, \mathbf{a}) = f(x, a, b, c) = a + bx + \frac{c}{x}$$

Least Squares Regression

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i	x_i	y_i
0	1	2.2
1	2	2.8
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Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}, c^{(0)}) = (1, 1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

Least Squares Regression

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i	x_i	y_i
0	1	2.2
1	2	2.8
2	3	3.6
3	4	4.5
$n = 4$	5	5.5

$$y(x) = f(x, \mathbf{a}) = f(x, a, b, c) = a + bx + \frac{c}{x}, m = 2$$

Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}, c^{(0)}) = (1, 1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

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$$\frac{\partial f}{\partial a} = 1, \quad \frac{\partial f}{\partial b} = x, \quad \frac{\partial f}{\partial c} = \frac{1}{x}$$

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$$\text{Tolerance: } \varepsilon_t = 10^{-6}$$

$$\frac{\partial f}{\partial a} = 1, \quad \frac{\partial f}{\partial b} = x, \quad \frac{\partial f}{\partial c} = \frac{1}{x}$$

$$\begin{bmatrix} \sum_{i=0}^4 \frac{\partial f}{\partial a} \cdot \frac{\partial f}{\partial a} & \sum_{i=0}^4 \frac{\partial f}{\partial a} \cdot \frac{\partial f}{\partial b} & \sum_{i=0}^4 \frac{\partial f}{\partial a} \cdot \frac{\partial f}{\partial c} \\ \sum_{i=0}^4 \frac{\partial f}{\partial b} \cdot \frac{\partial f}{\partial a} & \sum_{i=0}^4 \frac{\partial f}{\partial b} \cdot \frac{\partial f}{\partial b} & \sum_{i=0}^4 \frac{\partial f}{\partial b} \cdot \frac{\partial f}{\partial c} \\ \sum_{i=0}^4 \frac{\partial f}{\partial c} \cdot \frac{\partial f}{\partial a} & \sum_{i=0}^4 \frac{\partial f}{\partial c} \cdot \frac{\partial f}{\partial b} & \sum_{i=0}^4 \frac{\partial f}{\partial c} \cdot \frac{\partial f}{\partial c} \end{bmatrix} \begin{Bmatrix} \Delta a \\ \Delta b \\ \Delta c \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot \frac{\partial f}{\partial a} \\ \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot \frac{\partial f}{\partial b} \\ \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot \frac{\partial f}{\partial c} \end{Bmatrix}$$

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Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}, c^{(0)}) = (1, 1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

$$\frac{\partial f}{\partial a} = 1, \quad \frac{\partial f}{\partial b} = x, \quad \frac{\partial f}{\partial c} = \frac{1}{x}$$

$n = 4$

$$\begin{bmatrix} \sum_{i=0}^4 (1) \cdot (1) & \sum_{i=0}^4 (1) \cdot (x_i) & \sum_{i=0}^4 (1) \cdot \left(\frac{1}{x_i}\right) \\ \sum_{i=0}^4 (x_i) \cdot (1) & \sum_{i=0}^4 (x_i) \cdot (x_i) & \sum_{i=0}^4 (x_i) \cdot \left(\frac{1}{x_i}\right) \\ \sum_{i=0}^4 \left(\frac{1}{x_i}\right) \cdot (1) & \sum_{i=0}^4 \left(\frac{1}{x_i}\right) \cdot (x_i) & \sum_{i=0}^4 \left(\frac{1}{x_i}\right) \cdot \left(\frac{1}{x_i}\right) \end{bmatrix} \begin{Bmatrix} \Delta a \\ \Delta b \\ \Delta c \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot (1) \\ \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot (x_i) \\ \sum_{i=0}^4 [y_i - a - bx_i - \frac{c}{x_i}] \cdot \left(\frac{1}{x_i}\right) \end{Bmatrix}$$

Least Squares Regression

Example 04-09: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
3	3	2.4
4	4	2.0

$$y(x) = f(x, \mathbf{a}) = f(x, a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2$$

Least Squares Regression

Example 04-09: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
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$$y(x) = f(x, \mathbf{a}) = f(x, a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2$$

Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}) = (1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

Least Squares Regression

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$$y(x) = f(x, \mathbf{a}) = f(x, a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2, m = 1$$

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Tolerance: $\varepsilon_t = 10^{-6}$

$$\frac{\partial f}{\partial a} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(\frac{1}{b \cdot \sqrt{x}} \right), \quad \frac{\partial f}{\partial b} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(-\frac{a + \sqrt{x}}{b^2 \cdot \sqrt{x}} \right)$$

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$$y(x) = f(x, \mathbf{a}) = f(x, a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2, \text{ } m = 1$$

Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}) = (1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

$$\frac{\partial f}{\partial a} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(\frac{1}{b \cdot \sqrt{x}} \right), \quad \frac{\partial f}{\partial b} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(-\frac{a + \sqrt{x}}{b^2 \cdot \sqrt{x}} \right)$$

$$\begin{bmatrix} \sum_{i=0}^4 \frac{\partial f}{\partial a} \cdot \frac{\partial f}{\partial a} & \sum_{i=0}^4 \frac{\partial f}{\partial a} \cdot \frac{\partial f}{\partial b} \\ \sum_{i=0}^4 \frac{\partial f}{\partial b} \cdot \frac{\partial f}{\partial a} & \sum_{i=0}^4 \frac{\partial f}{\partial b} \cdot \frac{\partial f}{\partial b} \end{bmatrix} \begin{Bmatrix} \Delta a \\ \Delta b \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^4 \left[y_i - \left(\frac{a + \sqrt{x_i}}{b \cdot \sqrt{x_i}} \right)^2 \right] \cdot \frac{\partial f}{\partial a} \\ \sum_{i=0}^4 \left[y_i - \left(\frac{a + \sqrt{x_i}}{b \cdot \sqrt{x_i}} \right)^2 \right] \cdot \frac{\partial f}{\partial b} \end{Bmatrix}$$

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Initial Guess: $\mathbf{a}^{(0)} = (a^{(0)}, b^{(0)}) = (1, 1)$

Tolerance: $\varepsilon_t = 10^{-6}$

$$\frac{\partial f}{\partial a} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(\frac{1}{b \cdot \sqrt{x}} \right), \quad \frac{\partial f}{\partial b} = 2 \cdot \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right) \cdot \left(-\frac{a + \sqrt{x}}{b^2 \cdot \sqrt{x}} \right)$$

$$\begin{bmatrix} \sum_{i=0}^4 \left(2 \cdot \frac{a + \sqrt{x_i}}{b^2 \cdot x_i} \right) \cdot \left(2 \cdot \frac{a + \sqrt{x_i}}{b^2 \cdot x_i} \right) & \sum_{i=0}^4 \left(2 \cdot \frac{a + \sqrt{x_i}}{b^2 \cdot x_i} \right) \cdot \left[-2 \cdot \frac{(a + \sqrt{x_i})^2}{b^3 \cdot x_i} \right] \\ \sum_{i=0}^4 \left[-2 \cdot \frac{(a + \sqrt{x_i})^2}{b^3 \cdot x_i} \right] \cdot \left(2 \cdot \frac{a + \sqrt{x_i}}{b^2 \cdot x_i} \right) & \sum_{i=0}^4 \left[-2 \cdot \frac{(a + \sqrt{x_i})^2}{b^3 \cdot x_i} \right] \cdot \left[-2 \cdot \frac{(a + \sqrt{x_i})^2}{b^3 \cdot x_i} \right] \end{bmatrix} \begin{Bmatrix} \Delta a \\ \Delta b \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^4 \left[y_i - \left(\frac{a + \sqrt{x_i}}{b \cdot \sqrt{x_i}} \right)^2 \right] \cdot \left(2 \cdot \frac{a + \sqrt{x_i}}{b^2 \cdot x_i} \right) \\ \sum_{i=0}^4 \left[y_i - \left(\frac{a + \sqrt{x_i}}{b \cdot \sqrt{x_i}} \right)^2 \right] \cdot \left[-2 \cdot \frac{(a + \sqrt{x_i})^2}{b^3 \cdot x_i} \right] \end{Bmatrix}$$

Least Squares Regression

Least Squares Regression

Example 04-10: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
3	3	2.4
4	4	2.0

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2$

Least Squares Regression

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0	0.5	10.4
1	1	5.8
2	2	3.3
3	3	2.4
4	4	2.0

$n = 4$

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2$, $m = 1$

Least Squares Regression

Example 04-10: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	0.5	10.4
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2	2	3.3
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$n = 4$	4	2.0

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2, m = 1$

Linear Model: $Y(x) = \sqrt{y(x)} = g(A, B) = \frac{A}{\sqrt{x}} + B, m = 1$

Least Squares Regression

Example 04-10: Find least squares regression for the following data with the regression model

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
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$n = 4$	4	2.0

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2$, $m = 1$

Linear Model: $Y(x) = \sqrt{y(x)} = g(A, B) = \frac{A}{\sqrt{x}} + B$, $m = 1$

$$y(x) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}} \right)^2$$
$$\sqrt{y(x)} = \frac{a + \sqrt{x}}{b \cdot \sqrt{x}} = \frac{a}{b} \cdot \frac{1}{\sqrt{x}} + \frac{1}{b} = \frac{A}{\sqrt{x}} + B$$

Least Squares Regression

Example 04-10: Find least squares regression for the following data with the regression model

$n = 4$

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
3	3	2.4
4	4	2.0

i	x_i	$Y_i = \sqrt{y_i}$
0	0.5	3.224903
1	1	2.408319
2	2	1.816590
3	3	1.549193
4	4	1.414214

Nonlinear Model: $y(x) = f(a,b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2, m = 1$

Linear Model: $Y(x) = \sqrt{y(x)} = g(A,B) = \frac{A}{\sqrt{x}} + B, m = 1$

$$y(x) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2$$
$$\sqrt{y(x)} = \frac{a + \sqrt{x}}{b \cdot \sqrt{x}} = \frac{a}{b} \cdot \frac{1}{\sqrt{x}} + \frac{1}{b} = \frac{A}{\sqrt{x}} + B$$

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2	2	3.3
3	3	2.4
4	4	2.0

$n = 4$

i	x_i	$Y_i = \sqrt{y_i}$
0	0.5	3.224903
1	1	2.408319
2	2	1.816590
3	3	1.549193
4	4	1.414214

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2, m = 1$

Linear Model: $Y(x) = \sqrt{y(x)} = g(A, B) = \frac{A}{\sqrt{x}} + B, m = 1$

$$\phi_0(x) = \frac{1}{\sqrt{x}}, \phi_1(x) = 1$$

$$y(x) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2$$
$$\sqrt{y(x)} = \frac{a + \sqrt{x}}{b \cdot \sqrt{x}} = \frac{a}{b} \cdot \frac{1}{\sqrt{x}} + \frac{1}{b} = \frac{A}{\sqrt{x}} + B$$

Least Squares Regression

Example 04-10: Find least squares regression for the following data with the regression model

$n = 4$

i	x_i	y_i
0	0.5	10.4
1	1	5.8
2	2	3.3
3	3	2.4
4	4	2.0

$n = 4$

i	x_i	$Y_i = \sqrt{y_i}$
0	0.5	3.224903
1	1	2.408319
2	2	1.816590
3	3	1.549193
4	4	1.414214

Nonlinear Model: $y(x) = f(a, b) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2, m = 1$

Linear Model: $Y(x) = \sqrt{y(x)} = g(A, B) = \frac{A}{\sqrt{x}} + B, m = 1$

$\phi_0(x) = \frac{1}{\sqrt{x}}, \phi_1(x) = 1$

$$y(x) = \left(\frac{a + \sqrt{x}}{b \cdot \sqrt{x}}\right)^2$$
$$\sqrt{y(x)} = \frac{a + \sqrt{x}}{b \cdot \sqrt{x}} = \frac{a}{b} \cdot \frac{1}{\sqrt{x}} + \frac{1}{b} = \frac{A}{\sqrt{x}} + B$$

$$\begin{bmatrix} \sum_{i=0}^4 \phi_0(x_i) \cdot \phi_0(x_i) & \sum_{i=0}^4 \phi_0(x_i) \cdot \phi_1(x_i) \\ \sum_{i=0}^4 \phi_1(x_i) \cdot \phi_0(x_i) & \sum_{i=0}^4 \phi_1(x_i) \cdot \phi_1(x_i) \end{bmatrix} \begin{Bmatrix} A \\ B \end{Bmatrix} = \begin{Bmatrix} \sum_{i=0}^4 \phi_0(x_i) \cdot \sqrt{y_i} \\ \sum_{i=0}^4 \phi_1(x_i) \cdot \sqrt{y_i} \end{Bmatrix}$$